Experiment - 1

STUDY OF VARIOUS NETWORK COMMANDS USED IN LINUX AND WINDOWS

AIM: Study of various Network commands used in Linux And Windows.

Basic Networking commands:

arp -a : ARP is short form of address resolution protocol. Shows IP & MAC addres of your PC and router.
hostname: displays the name of your computer.
ipconfig/all - Show detailed TCP/IP config
nbstat -a - helps resolve NetBIOS names
netstat -r - Shows active TCP/IP connections, routing table, and interfaces stats.
nslookup - performs DNS lookup (IP, MX, NS records)
pathping: combines ping + tracert
ping localhost: tests connectivity using ICMP.
route - Shows/edits the IP routing table.

Important Linux networking commands:

ifconfig/ip addr - Shows IP address, MAC, subnet mask of interfaces.
ping - tests connectivity using ICMP
traceroute - Shows the hop-by-hop path to a destination

netstat -tulnp - shows active connections and listening ports.

ss - modern replacement for netstat.

nslookup/dig - performs DNS lookups.

arp -a - shows active connections and listening ports.

route -n/ip route - displays the system's hostname.

whois - gets domain registration info.

Procedure:

1. Open Command Prompt for Terminal.
2. Run ping <hostname/IP> to check connectivity to a host.
3. Run ipconfig or ifconfig / ip addr to view IP configuration
4. Run tracert <host> or traceroute <host> to trace the path to a destination.
5. Run netstat -a to view active connections and listening ports.
6. Run nslookup <domain> to resolve a domain name to its IP address.
7. Run arp -a to view the ARP cache.
8. Run route print or route -n to view the routing table.
9. Record the output.

8/10

Result:

Thus, the various network commands in Linux and Windows were studied and executed successfully.